



Jothish Kumar P 251801123-2025-AIDS ▾

J2

Started on Thursday, 6 August 2026, 11:11 AM**State** Finished**Completed on** Thursday, 6 August 2026, 11:16 AM**Time taken** 5 mins 8 secs**Marks** 1.00/1.00**Grade** 10.00 out of 10.00 (100%)

Question 1 | Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using counter method.

```
void reverse(int n)
{
    int rev = 0, remainder;
    while (n != 0)
    {
        remainder = n % 10;
        rev = rev * 10 + remainder;
        n /= 10;
    }
    print(rev);
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer:

```
1  #include <stdio.h>
2
3  int main()
4  {
5      int n, rev = 0, remainder;
6      int count = 3;
7
8      scanf("%d",&n);
9
10     while(n != 0)
11     {
12         count++;
13
14         remainder = n % 10;
15         count++;
16
17         rev = rev * 10 + remainder;
18         count++;
19
20         n = n / 10;
21         count++;
22     }
23
24     printf("%d", count);
25
26     return 0;
27 }
```

	Input	Expected	Got	
✓	12	11	11	✓

	Input	Expected	Got	
✓	1234	19	19	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

[Back to Course](#)



Jothish Kumar P 251801123-2025-AIDS ▾

J2

Started on Thursday, 6 August 2026, 11:04 AM**State** Finished**Completed on** Thursday, 6 August 2026, 11:09 AM**Time taken** 4 mins 40 secs**Marks** 1.00/1.00**Grade** 10.00 out of 10.00 (100%)

Question 1 | Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time

complexity using counter method.

```
void function(int n)
{
    int c = 0;
    for(int i=n/2; i<n; i++)
        for(int j=1; j<n; j = 2 * j)
            for(int k=1; k<n; k = k * 2)
                c++;
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer:

```
1  #include<stdio.h>
2
3  int main()
4  {
5      int n;
6      int c = 0;
7
8      scanf("%d",&n);
9
10     int count = 1;
11     for(int i=n/2; i<n; i++)
12     {
13         count++;
14
15         for(int j=1; j<n; j=j*2)
16         {
17             count++;
18
19             for(int k=1; k<n; k=k*2)
20             {
21                 count++;
22                 c++;
23                 count++;
24             }
25
26             count++;
27         }
28
29         count++;
30     }
31
32     count++;
33
34     printf("%d",count);
35
36     return 0;
37 }
```

	Input	Expected	Got	
✓	4	30	30	✓
✓	10	212	212	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

[Back to Course](#)





Jothish Kumar P 251801123-2025-AIDS ▾

J2

Started on Thursday, 6 August 2026, 10:53 AM**State** Finished**Completed on** Thursday, 6 August 2026, 10:57 AM**Time taken** 4 mins 8 secs**Marks** 1.00/1.00**Grade** 10.00 out of 10.00 (100%)

Question 1 | Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using counter method.

```
Factor(num) {
{
    for (i = 1; i <= num; ++i)
    {
        if (num % i == 0)
        {
            printf("%d ", i);
        }
    }
}
```

Note: No need of counter increment for declarations and scanf() and counter variable printf() statement.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer:

```
1  #include<stdio.h>
2  int main(){
3      int n;
4      scanf("%d",&n);
5      int c=0;
6      c++;
7  for(int i=1;i<=n;i++){
8      c++;
9      if(n%i==0){
10         c++;
11     }
12     c++;
13 }
14 printf("%d",c);
15 }
```

	Input	Expected	Got	
✓	12	31	31	✓
✓	25	54	54	✓
✓	4	12	12	✓

Passed all tests! 

Correct

Marks for this submission: 1.00/1.00.

[Back to Course](#)



Jothish Kumar P 251801123-2025-AIDS ▾

J2

Started on Thursday, 6 August 2026, 10:40 AM**State** Finished**Completed on** Thursday, 6 August 2026, 10:45 AM**Time taken** 5 mins 20 secs**Marks** 1.00/1.00**Grade** 10.00 out of 10.00 (100%)

Question 1 | Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using the counter method.

```
void func(int n)
{
    if(n==1)
    {
        printf("*");
    }
    else
    {
        for(int i=1; i<=n; i++)
        {
            for(int j=1; j<=n; j++)
            {
                printf("*");
                printf("*");
                break;
            }
        }
    }
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer: (penalty regime: 0 %)

```
1  #include<stdio.h>
2
3  int main()
4  {
5      int n;
6      int count = 0;
7
8      scanf("%d",&n);
9
10     if(n==1)
11     {
12         printf("*");
13     }
14     else
15     {
16         int i,j;
17
18         count += 1;
19
20         for(i=1; i<=n; i++)
21         {
22             count += 1;
23             count += 1;
24
25             for(j=1; j<=n; j++)
26             {
27                 count += 1;
28             }
29         }
30     }
```

```
29         break;
30     }
31
32     count += 1;
33
34     count += 1;
35 }
36
37 count += 1;
38 }
39
40 printf("%d",count);
41
42 return 0;
43 }
```

	Input	Expected	Got	
✓	2	12	12	✓
✓	1000	5002	5002	✓
✓	143	717	717	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

[Back to Course](#)



Jothish Kumar P 251801123-2025-AIDS ▾

J2

Started on Thursday, 6 August 2026, 10:11 AM**State** Finished**Completed on** Thursday, 6 August 2026, 10:13 AM**Time taken** 1 min 53 secs**Marks** 1.00/1.00**Grade** 10.00 out of 10.00 (100%)

Question 1 | Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using the counter method.

```
void function (int n)
```

```
{
```

```
    int i= 1;
```

```
    int s =1;
```

```
    while(s <= n)
```

```
    {
```

```
        i++;
```

```
        s += i;
```

```
    }
```

```
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

For example:

Input	Result
9	12

Answer: (penalty regime: 0 %)

```

1  #include <stdio.h>
2
3  int main()
4  {
5      int n;
6      scanf("%d", &n);
7
8      int counter = 0;
9      int i = 1;
10     int s = 1;
11
12     counter += 2;
13     while (1)
14     {
15         counter++;
16         if (!(s <= n))
17             break;
18
19         i++;
20         counter++;
21
22         s += i;
23         counter++;
24     }
25
26     printf("%d", counter);
27
28     return 0;
29 }
```

	Input	Expected	Got	
✓	9	12	12	✓
✓	4	9	9	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

[Back to Course](#)